

DESIGNING

the

THINKERS

of

TOMORROW

by

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HISTORY

figure 1 & 2

To look at where to place design's role in public education now, it is important to reflect backwards on how public school in America actually started, and how it got to where it is today. In 1647, the General Court of the Massachusetts Bay Colony put into law that every town of at least 50 families must have a public elementary school, and each town with at least 100 families must have a Latin school (Race Forward). The goal of these institutions was for the youth of the Puritan colonies to be able to read and understand the Bible (Race Forward). Now of course, the only people allowed to attend these schools were caucasian males (Great Homeschool Conventions).

It can be seen all throughout history that knowledge both was power and challenged power. Revolutions occurred when marginalized groups gained the ability to learn, enabling them to think freely, speak freely, and communicate their thoughts efficiently. Control of knowledge meant control of society. This reminds me of how Bruno Munari discusses designers creating for the elite upper class of consumers in his book *Design as Art*. Just as public education was long gatekept by race, class, and gender, Munari's designers served only those who could afford to be served. In both cases, the moment access began to widen, the systems that depended on exclusion started to crack.

In this early era of education, the most elite actually stuck to home schooling. Others used public schools, private schools, and schools that specialized in religious affiliations. However, school was still not tuition free, and each parent who wanted to educate their children had to pay (Great Homeschool Conventions). Up until the 1800s this system remained, and only white men were in public schools

(Kew). There were some public girl's schools, but their education was not nearly the same, as they were mainly taught housewife skills and literacy for bible reading (Great Homeschool Conventions). Regardless of opposition from wealthy wage earners who did not want to contribute their tax dollars, the first public school in the United States, Boston English, opened in 1820 (Race Forward). Instead of teaching Greek and Latin, it shifted education to focus on practical topics like business, math, geography, navigation, and logic (Great Homeschool Conventions). Most importantly, it was free and open to all white boys, allowing for social mobility (Kew). Then in 1832, Horace Mann became the secretary of education in Massachusetts (Great Homeschool Conventions). He believed that education was a natural right (Kew), and therefore spearheaded the first laws that made attendance compulsory, officially passed in 1852 (Great Homeschool Conventions).

While this seems like positive social reform, it can also be viewed as a way of ensuring that the flood of immigrant children to the new country were "civilized" (Race Forward). Within the 1830s, about 55% of children aged 5 to 14 were enrolled in public schools (Kober & Rentner). By 1870, this proportion had risen to about 78% (Kober & Rentner). Also in the 1830s, many laws in the south were passed forbidding slaves to be educated, but even still 5% became so at great risk (Race Forward).

What kind of contexts and breaks in world history have acted on it?

What does the historic evolution of your topic look like?

Horace Mann also strongly advocated for the Prussian model of education, which spread greatly throughout schools throughout the 19th century and is still highly modeled in public schools today (Kew). In 18th century Prussia, education shifted from creating critical thinkers to compliant soldiers (Cardinal Institute). This culture change was due to the exposure of their military's weaknesses after Napoleon's invasion. Citizens of Prussia were unreliable and unloyal, so Prussian leaders implemented new schooling methods to change this (Cardinal Institute). Horace Mann visited Prussia and took the ideals of making orderly citizens back with him (Cardinal Institute). Congress understood that schools could not immediately keep up to new regulations set by them, so the government started issuing standards (Kew). Schools began operating like conveyer belts: organized by age, taught the same subjects, and ranked by standardized metrics (Kew).

Starting in the 1850s and going into the 20th century, secondary schools became both more popular and more necessary. Especially in urban areas, public schools offered a solution to older children being let loose to wreak havoc amongst new and growing cities (Kew). However, these institutions were still incredibly divisive. Mixed-gender schools only appeared in the late 18th-early 19th century. Franklin and Marshall College in Pennsylvania opened first as a coed secondary school in 1787, and in 1837 Oberlin became the first college to accept women (Robertson). After the Civil War (1861-1865), white supremacists in southern state governments passed the "Jim Crow" laws, enforcing segregation in public schools and other major spaces of life (Kober & Rentner).

The legal infrastructure of segregation was further cemented in 1896, when the Supreme Court ruled in *Plessy v. Ferguson* that "separate but equal" facilities were constitutional, effectively encoding racial exclusion into the design of public schooling itself (Kober & Rentner). Black students were funneled into chronically underfunded schools with outdated materials and undertrained teachers, while the law looked the other way (Race Forward). By the early 1900s, a growing counter-movement began to challenge the Prussian conveyer belt. Philosopher and educator John Dewey argued that schools should cultivate curiosity, critical thinking, and democratic participation rather than produce obedient citizens (Kew). His progressive ideas introduced concepts like project-based learning and student-centered classrooms, many of which still circulate in education reform conversations today (Kew). However, Dewey's influence was uneven, and the standardized, compliance-driven structure of American public schooling proved deeply resistant to change (Cardinal Institute).

It was not until 1954, with the Supreme Court's ruling in *Brown v. Board of Education*, that the embedded laws of segregation were formally dismantled (Race Forward). The decision declared that separate educational facilities were inherently unequal, marking one of the most significant breaks in the history of American public education (Kober & Rentner). And yet the ruling was only the beginning. The work of actually designing an equitable school system, one accessible to all children regardless of race, class, or gender, remained very much unfinished.



figure 3

What did the first design idea look like?

INTO THE NOW

The work left unfinished by *Brown v. The Board of Education* embedded itself into how schools are built, equipped, and run today. The current moment in public education is defined by a wave of design interventions: some architectural, some technological, all promising to fix what exclusion broke. However, the question of who these interventions actually serve remains.

Walk into a typical American public school building and the Prussian blueprint is still visible: rows of desks facing a board, fluorescent lighting, corridors timed by bells, and rooms organized by age. The physical environment has not changed nearly as much as the rhetoric around it suggests, and yet there is growing evidence that it should. Research has shown that the design of a learning space accounts for as much as 10 to 15 percent of variance in academic outcomes, with factors like natural lighting, acoustics, air quality, and spatial flexibility all measurably affecting how well students learn (EDspaces). Natural light alone has been shown to increase test scores and reduce absenteeism, while poor acoustics actively impair reading comprehension (EDspaces). This is where Don Norman's concept of affordances, outlined

in his book *The Design of Everyday Things*, becomes a lens. Norman argues that well-designed objects communicate their own use intuitively without instruction. A chair pulled into a circle signals conversation. A row of chairs facing a stage signals passivity. The same logic applies at architectural scale: a classroom designed with fixed desks and a single focal point at the front of the room is not a neutral container. It is a set of instructions, telling students to sit still, face forward, receive information, and not collaborate. The physical environment of a school is "a silent curriculum" that shapes behavior, interaction, and learning culture (EDspaces). Progressive redesigns are pushing back against this. Imagine a classroom with flexible furniture that reconfigures from lecture to workshop to small group. This room has writable walls, varied seating that accommodates different bodies and working styles, and connections to natural light and outdoor space. This ideal learning environment encompasses what researchers now call salutogenic, or health-promoting, learning environments (Minhas). These spaces are being built, but largely in well-funded districts where design innovation is a line item in the budget. The redesign of learning spaces is

happening, and the research behind it is strong. The problem is that it is not happening everywhere. The same uneven pattern defines the technology conversation. The past decade has seen an enormous push to get devices into students' hands, and in terms of raw access (laptops, iPads, etc.), significant ground has been covered. But access to a laptop is not the same as access to meaningful learning. Researchers have identified what they call a "digital use divide," the gap between students who use technology actively and creatively and those who use it only to passively complete assignments (K-12 Dive). Closing the hardware gap did not close the learning gap, it just reframed it. Now generative AI is accelerating that divide. 86 percent of students and 85 percent of teachers were already using AI tools during the 2024 to 2025 school year, yet only 18 percent of principals had received any formal guidance on how to lead that shift (Educational Innovation 360). AI is everywhere in schools and almost nowhere in policy. The result is an environment where teachers and students are navigating these tools on their own without frameworks, training, or shared standards for what responsible use looks like.



↶ figure 5

For students in well-resourced schools, this means access to AI tutoring tools, writing assistants, and personalized feedback systems that adapt to their pace and learning style. For students in underfunded districts, it often means none of that, or worse, exposure to poorly designed algorithmic tools that were not built with them in mind. A 2024 report from the U.S. Commission on Civil Rights found that AI in education has the potential to both narrow opportunity gaps and deepen the digital divide; its civil rights implications must be evaluated seriously before implementation (U.S. Commission on Civil Rights). Well-governed AI systems can boost learning and narrow existing gaps, but poorly designed tools can deepen bias and eliminate trust, all while the long-term effects on students' critical thinking and creativity remain uncertain (Brookings).

The design world has produced research-backed innovations in both physical space and technology. Flexible classrooms work. AI tutoring tools (when implemented well) show meaningful learning gains. The problem is not a shortage of good ideas. It actually has not changed since the original public schools in the 1800s: access to good design has always been distributed along existing lines of wealth and race. The gap *Brown v. The Board of Education* promised to close did not close, but rather found new forms in socioeconomic and racial disparity.

I personally was pumped through the public education system not too long ago. The most notable classes from my high school experience were the ones where I interacted most with my peers and participated heavily in project based learning. It is no surprise that those classes were my art and computer science ones. I firmly believe that my love for these two subjects was not simply natural, but manufactured from how hands-on the learning involved with them was. I am now a Computer Science and Design combined major at a global research university. I am newly able to observe the public education world from the other side, learning the design behind the very system that raised me. Through Northeastern, I tutor at an after school program for K-12 graders called Boston 826. I see how the system acts in real time. The classroom I tutor in is open concept with a big library, soft cushioned seats, and tables that face each other and facilitate conversation. However, I have also seen how reliant students are becoming on AI tools to do the learning for them instead of being a guide along the way, illustrating a growing gap between youth and critical thinking. Public schools now more than ever need to raise generations that ask questions, not regurgitate information. The physical and technological design of a school can either build that capacity or erase it. I want to spend time in my own practice as a designer navigating that tension, and point the future of education design towards opening the floor to more questions.



↷ figure 6

TOMORROW

What kind of world are we designing schools to produce? This question is where the future of public education lies. Innovations emerging right now are exciting and promising, but whether they serve the many or the few will depend entirely on the values embedded in their design.

AI tutoring systems are transforming the Prussian conveyor belt system by meeting students where they actively are. Some schools already utilize models where AI tutors handle personalized skill-building in the morning, completing core academics in roughly two hours, while teachers spend the rest of the day leading workshops, mentorship sessions, and collaborative projects (Hunt Institute). The teachers in this process become guides rather than lecturers, exactly what John Dewey argued for. Other research has shown that the most effective AI tutoring model is a human-AI hybrid, where technology handles content delivery and humans handle the work of critical thinking, discussion, and relationship-building (Brookings). New technologies like mixed reality push the boundaries of education even further. Over 40 percent of K-12 schools are expected to incorporate AR and VR technologies, with studies showing that students in VR environments report feeling significantly more emotionally connected to content than those in traditional classrooms (Reframe XR). These tools further learning by putting the course content literally in the learners' hands. For instance, a student who struggles to visualize molecular structures can hold them in an AR environment, having the lesson develop in their palm. Classrooms are beginning to expand beyond their walls. Though AI and immersive technology will never replace high-quality human teaching, the most successful implementations will be the ones that enhance educators rather than sideline them (WEF).

This immersive technology has benefits in theory, but it must be designed with equity in mind, fostering opportunity across all students regardless of geography, income, or ability. The Prussian model of education is one of the most durable design monocultures in human history. In his *Designs for the Pluriverse*, Arturo Escobar describes such monocultures as being produced by designers serving capitalist and modernizing ends, producing a single vision of how the world should be organized. Every child follows one standard curriculum, structure, and metric of success. A pluriversal approach to education design needs to reject that, especially when it comes to designing new learning tools. A pluriversal school system would mean that every community holds its own knowledge, ways of learning, and vision of what it means to succeed. Schools should be designed to draw those out rather than standardize them away.

Donna Haraway's concept of making kin, developed in *Staying with the Trouble*, offers another frame for what schools could be. Haraway argues that building livable futures requires what she calls sym-poiesis, or making-with, as opposed to self-making in isolation. She argues that living well together demands multiracial, multi-kinded relationships built on interdependence. Schools are one of the only institutions in society that bring people across lines of difference together by design. They are, or could be, the infrastructure for making kin if we design them in such a way.

This is the direction that researchers like Professor Nabeel Gillani at Northeastern University are already working toward. Gillani's Plural Connections Group uses computation and design to foster pluralism as an active, inclusive response to diversity, applying data science and design specifically to address educational inequalities rooted in segregated schools and neighborhoods (Gillani). His current research observes that diverse schools do not guarantee diverse friendships. Even in schools that appear integrated on paper, students self-segregate along socioeconomic lines, driven by forces like curricular tracking and social clustering that no attendance policy change can alone fix (Gillani, 2026). To address this, his lab is developing a Youth Civic Leadership Fellowship. This would be a student-driven program where young people from different socioeconomic backgrounds collaboratively research and design interventions for within-school segregation, studying the theory that cross-difference relationships are built through shared civic purpose rather than proximity (Gillani, 2026). This project combines Haraway's making-with with Escobar's vision of communities designing themselves.

figure 8

My vision
of education in the future is a school that
functions like a living, breathing system. It is
physically flexible and designed to signal collaboration
rather than compliance, technologically powered
but human-led, and pluriversal in its understanding of
what knowledge looks like and whose knowledge counts.
Through ethically, empathetically, critically, and open-
mindedly designing schools, society has the power to raise
generations that ask questions, build relationships, and
take responsibility for the world they are inheriting.

figure 9

SPECULATIVE QUESTION

If public education is a product of capitalist design, then it has the potential to be redesigned. How can we reimagine existing systems to produce curiosity, empathy, open-mindedness, and genuine academic interest? What tools help this, and what tools hurt it?

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